

Homeworld 1

Exercise 1

```
In[1]:= (* Func draws from the uniform distribution on [a,b],
        using a standard uniform sample given by the function RandomReal[] *)
        drawUniformNumber[a_, b_] := a + (b - a) * RandomReal[];
        reelle Zufallszahl

In[2]:= (* Func draws n samples of U=(U1, U2) w/ U1~U[-1,1], U2~U[0,1] *)
        drawTwoSamples[n_] :=
        Table[{drawUniformNumber[-1, 1], drawUniformNumber[0, 1]}, {i, 1, n}];
        Tabelle

In[3]:= (* Computes numerically given sum of n summands *)
        n = 100 000;
        U = drawTwoSamples[n];
        4 / n Sum[If[U[[i]][[1]]^2 + U[[i]][[2]]^2 ≤ 1, 1, 0], {i, 1, n}]
        s... wenn

Out[5]= 78 477
        25 000
```

Exercise 2

```
In[6]:= (* Func draws from the Weibull distr. w/ shape param. beta_ and scale param. 1 *)
        drawWeibullNumber[beta_] := (-Log[RandomReal[]])^(1/beta)
        L... reelle Zufallszahl
```

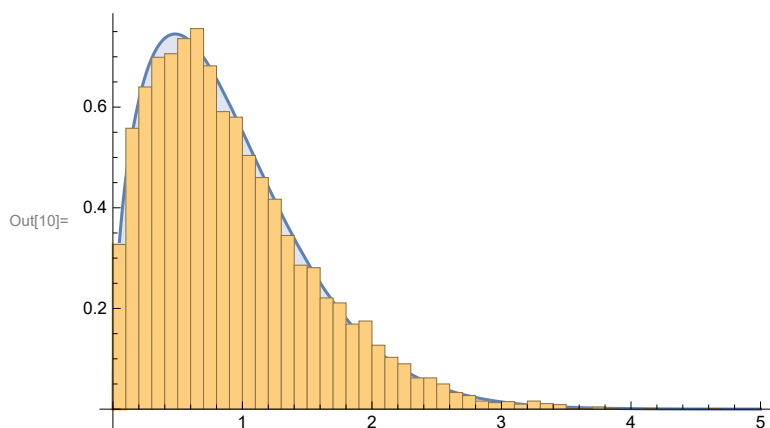
```

In[7]:= (* Plots a histogram of n simulated values of the Weibull
         distr. against the PDF of the Weibull distr. w/ shape param. beta *)
         [Wahrscheinlichkeitsdichtefunktion]

beta = 1.5;
n = 10000;
data = Table[drawWeibullNumber[beta], n];
         [Tabelle]

Show[
         [zeige an]
         Plot[PDF[WeibullDistribution[beta, 1], x] // Evaluate, {x, 0, 5}, Filling -> Axis],
         [stell... [W... [Weibull-Verteilung] [werte aus] [Füllung] [Achse]
         Histogram[data, Automatic, "PDF"]]
         [Histogramm] [automatisch] [Wahrscheinlichkeitsdichtefunktion]

```



Exercise 4

```

In[11]:= (* Find maximum of func z(x) *)
         [finde]

z[x_] := E / Sqrt[2 * Pi] * x^(-3/2) * Exp[-1 / (2 * x)];
         [... [Quadratw... [Kreiszahl π] [Exponentialfunktion]

c = Maximize[{z[x], x > 0}, x] [[1]]
         [maximiere]

```

Out[12]= $3 \sqrt{\frac{3}{2 e \pi}}$

```

In[13]:= (* Generates n samples from inverse Gaussian using A-R *)
f[x_] := z[x] Exp[-x/2];
           |Exponentialfunktion
g[x_] := (1/2) * Exp[-x/2];
           |Exponentialfunktion

n = 10000;
X = {};

While[Length[X] < n,
  |solange |Länge

  (* samples from E(1/2) using Inversion *)
  |Exponentialkonstante E
  Y = -2 * Log[RandomReal[]];
  |L... |reelle Zufallszahl
  If[RandomReal[] <= f[Y] / (2 * c * g[Y]), AppendTo[X, Y]; ] ;
  |w... |reelle Zufallszahl |hänge an bei
  ];

In[18]:= (* Constructs confidence intervals *)
XSquared = X^2;
meanX = Mean[X];
           |arithmetisches Mittel
meanXSquared = Mean[XSquared];
           |arithmetisches Mittel
varX = Variance[X];
           |Varianz
varXSquared = Variance[XSquared];
           |Varianz

Print["Sample mean of X is: ", meanX]
|gib aus
Print["Sample mean of X^2 is: ", meanXSquared]
|gib aus
Print["(Unbiased) sample variance of X is: ", varX]
|gib aus
Print["(Unbiased) sample variance of X^2 is: ", varXSquared]
|gib aus

Sample mean of X is: 0.995353
Sample mean of X^2 is: 1.98367
(Unbiased) sample variance of X is: 0.993044
(Unbiased) sample variance of X^2 is: 33.8872

```

```
In[27]:= (* Confidence interval for EX^2 *)
criticalTValue = Quantile[StudentTDistribution[n - 1], 0.975];
               |Quantil      |Student-t-Verteilung
meanXSquared - criticalTValue * varXSquared / Sqrt[n]
               |Quadratwurzel
meanXSquared + criticalTValue * varXSquared / Sqrt[n]
               |Quadratwurzel
```

Out[28]= 1.31942

Out[29]= 2.64793

```
In[30]:= (* Confidence interval for VarX *)
```

```
In[31]:= criticalChiSquaredValueLower = Quantile[ChiSquareDistribution[n - 1], 0.025];
               |Quantil      |Chi-Quadrat-Verteilung
criticalChiSquaredValueUpper = Quantile[ChiSquareDistribution[n - 1], 0.975];
               |Quantil      |Chi-Quadrat-Verteilung
varX * (n - 1) / criticalChiSquaredValueUpper
varX * (n - 1) / criticalChiSquaredValueLower
```

Out[33]= 0.966082

Out[34]= 1.02116